

Ex No: 1

IMPLEMENT EIGHT QUEENS PROBLEM

Aim:

To Implement Eight Queens Problem using backtracking

Procedure:

Program

N = 8 # Size of the chessboard (8x8)

Function to print the solution

def printSolution(board):

 for row in board:

 for i in range(N):

 print("Q" if row[i] else ".", end=" ")

 print()

 print() # Add a newline for readability

Function to check if a queen can be placed on board[row][col]

def isSafe(board, row, col):

 # Check the column

 for i in range(row):

 if board[i][col]:

 return False

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# Check the upper left diagonal
for i, j in zip(range(row - 1, -1, -1), range(col - 1, -1, -1)):
    if board[i][j]:
        return False

# Check the upper right diagonal
for i, j in zip(range(row - 1, -1, -1), range(col + 1, N)):
    if board[i][j]:
        return False
return True

# Function to solve the 8 Queens problem using backtracking
def solve(board, row):
    if row == N:
        printSolution(board)
        return True # Stop after finding one solution

    for col in range(N):
        if isSafe(board, row, col):
            board[row][col] = 1 # Place queen

            if solve(board, row + 1): # Recur to place next queen
                return True # Stop after first valid solution

            board[row][col] = 0 # Backtrack (remove queen)

    return False

# Main function to initialize the board and start solving the problem
def eightQueens():
    board = [[0 for _ in range(N)] for _ in range(N)]

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if solve(board, 0):
    print("One possible solution is:")
else:
    print("Solution does not exist")

# Calling the function
eightQueens()
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Output:

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One of the possible solutions is:
. . . . Q . . .
. . . Q . . . .
. . . . . Q . .
Q . . . . . . .
. . . . . Q .
. Q . . . . . .
. . Q . . . . .
. . . . . . Q
```

```

... Q . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . . .

```

One possible solution is: